

Zoo201 Current Papers Questions

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MCQS PART

1. A **statocyst** is a pit like invagination of the exoskeleton that contains setae and a group of cemented sand grains called a **statolith**.
2. Pycnogonids live on the ocean floor and frequently feed on cnidarian polyps and **ectoprocts**.
3. Crayfish have **supraesophageal** and **subesophageal** ganglia that receive sensory input from receptors in the head and control the head appendages.
4. The common marine mussel, **Mytilus**, uses the former method, while oysters employ the latter.
5. An insect that transmits a disease is known as a **vector**, and the disease is referred to as a **vector borne disease**.
6. Insects are responsible for the pollination of **approximately 65%** of all plant species.

Subjective PART

Lobopodium [Gk. lobes = lobe; podium = foot] (2 marks)

- It is a short, finger or tongue-like projection which is accompanied by a flow of endoplasm and ectoplasm.
- The pseudopodium is broad with rounded or blunt tips. The ectoplasmic area is distinctly clear, called the hyaline cap.
- It is the characteristic of many amoebas such as Amoeba, Chaos, Entamoeba, Arcella, Diffugia

Functions of Malpighian Tubules: (2 marks)

Malpighian tubules for excreting nitrogenous wastes. Malpighian tubules, absorb waste materials from the blood and empty them into the gut tract

Name the Animals which included in class Diplopoda? (2 marks)

Members of the class diplopods are Millipedes Most millipedes are slow-moving detritivores, eating decaying leaves and other dead plant matter

Holometabolous Metamorphosis (2 marks)

In holometabolous (Gr. holos, whole) metamorphosis, immature are called larvae because they are very different from the adult in body form, behavior, and habitat. • The number of larval instars is species specific, and the last larval molt forms the pupa.

Example: Beetles, butterflies and moths, flies, and wasps.

Paurometabolous metamorphosis?

Paurometabolous (Gr. pauros, small) metamorphosis involves a species-specific number of molts between egg and adult stages, during which immature gradually take on the adult form. Immatures are called nymphs.

Paurometabolous is the type of growth that occurs in insects such as cockroaches, which includes just three stages, i.e. egg, nymph, and adult. It is called incomplete metamorphosis or paurometabolous since it lacks the pupa stage.

Example: Grasshoppers and chinch bugs

Trepang

The body wall of sea cucumbers, when boiled and dried, is known as trepang in Asian countries.

Phylum Annelida (General Characteristics)

Characteristics of the phylum Annelida include:

1. Body metameric, bilaterally symmetrical, and wormlike.
2. Protostome characteristics include spiral cleavage, trochophore larvae (when larvae are present), and schizocoelous coelom formation.
3. Organ Level of organization.
4. Tube-within-tube body cavity.
5. Paired, epidermal setae.
6. Closed circulatory system.
7. Dorsal suprapharyngeal ganglia and ventral nerve cord(s) with ganglia.

8. Metanephridia (usually) or protonephridia.

Book lungs

- Some arachnids possess structures, called **book lungs**, that are assumed to be modifications of the book gills in the Merostomata.
- Book lungs are paired invaginations of the ventral body wall that fold into a series of leaflike lamellae.
- Air enters the book lung through a slitlike opening and circulates between lamellae.

Synchronous Flight Mechanism of Insects

- Direct or synchronous flight is a mode of transportation that is fueled by wing muscles that insert directly into the wing base.
- The insertion point of the wing is hinged which enables the muscles downward movements to lift the wing portion upward and upward movements pull the wing portion downward.

Butterflies, dragonflies, and grasshoppers are examples of insects with a synchronous flight mechanism.

Circulation or Circulatory system in Crayfish (5 marks)

- Circulation in crayfish is similar to that of most arthropods.
- Dorsal, anterior, and posterior arteries lead away from a muscular heart.
- Branches of these vessels empty into sinuses of the hemocoel.
- Blood returning to the heart collects in a ventral sinus and enters the gills before returning to the pericardial sinus, which surrounds the heart.
- Crustaceans have an open circulatory system meaning that all their blood is not contained within vessels, instead, blood is drawn in to the heart through holes called ostia, then pumped out again to circulate through the tissues and return again to the heart.

Defined Flukes (2 marks)

The approximately eight thousand species of parasitic flatworms in the class Trematoda (Gr. trematodes, perforated form) are collectively called flukes, which describes their wide, flat shape.

- Almost all adult flukes are parasites of vertebrates, whereas immature stages may be found in vertebrates or invertebrates, or encysted on plants.

Compound eyes (2 Marks)

- Compound eyes are well developed in most adult insects.
- They are similar in structure and function to those of other arthropods, although zoologists debate their possible homology (common ancestry) with those of crustaceans, horseshoe crabs, and trilobites.

- Compound eyes consist of a few to 28,000 receptors, called ommatidia that fuse into a multifaceted eye.
- The compound eye is better suited for detecting movement.

Class Crinoidea

Members of the class Crinoidea (Gr. krinon, lily ovides, in the form of) include the sea lilies and the feather stars. • They are the most primitive of all living echinoderms and are very different from any covered thus far.

- Approximately 630 species are living today; however, an extensive fossil record indicates that many more were present during the Paleozoic era, 200 to 600 million years ago.
- Crinoids are pentamerous, stalked echinoderms with a cuplike body bearing five usually branched and commonly featherlike arms.
- Most of a crinoid's body consists of an endoskeleton composed of numerous calcareous pieces, called plates or ossicles.

Autotomy of Ophiuroids or Regeneration in Ophiuroids

- Like sea stars, ophiuroids can regenerate lost arms. • If a brittle star is grasped by an arm, the contraction of certain muscles may sever and cast off the arm hence the common name brittle star. • This process, called **autotomy** (Gr. autos, self-tomos, to cut), is used in escape reactions.
- The ophiuroid later regenerates the arm.

What is the Function of Gut Tract in Arachnida?

Explains (5 marks)

The gut tract of arachnids is divided into three regions. • The anterior portion is the foregut, and the posterior portion is the hindgut. Both develop as infoldings of the body wall and are lined with cuticle. • A portion of the foregut is frequently modified into a pumping pharynx, and the hindgut is frequently a site of water reabsorption.

- The midgut between the foregut and hindgut is noncuticular and lined with secretory and absorptive cells.
- Lateral diverticula increase the area available for absorption and storage.

Crustacean or Crayfish nervous systems (5 marks)

- Crustacean nervous systems show trends similar to those in annelids and arachnids.
- Primitively, the ventral nervous system is ladderlike.
- Higher crustaceans show a tendency toward centralization and cephalization.
- Crayfish have supraesophageal and subesophageal ganglia that receive sensory input from receptors in the head and control the head appendages.
- The ventral nerves and segmental ganglia fuse, and giant neurons in the ventral nerve cord function in escape responses
- When nerve impulses are conducted posteriorly along giant nerve fibers of a crayfish, powerful abdominal flexor muscles of the abdomen contract

alternately with weaker extensor muscles, causing the abdomen to flex (the propulsive stroke) and then extend (the recovery stroke) • The telson and uropods form a paddle like “tail” that propels the crayfish posteriorly.

Mating Behavior in Insects (5 marks)

Mating behavior is typically viewed as comprising all events from pair formation through courtship to the final breakup of the mating pair. Locating and recognizing mates: - Chemical attractants - Humming sound - Courtship ritual mating: Internal fertilization by insertion of the male intromittent organ into the female genital tract for deposition of sperm is the usual method of copulation.

Most insects have complex mating behaviors for locating and recognizing a potential mate, for positioning a mate for copulation, or for pacifying an aggressive mate. Mating behavior may involve pheromones (moths, order Lepidoptera), visual signals (fireflies, order Coleoptera), and auditory signals (cicadas, order Homoptera; and grasshoppers, crickets, and katydids, order Orthoptera).

Bipinnaria larva & Brachiolaria larva (5 Marks)

Bipinnaria larva

After gastrulation, bands of cilia differentiate, and a bilaterally symmetrical larva, called a bipinnaria larva, forms. • The larva usually feeds on planktonic protists. •

Brachiolaria larva

• The development of larval arms results in a brachiolaria larva, which settles to the substrate, attaches, and metamorphoses into a juvenile sea star. • Later embryonic stages are ciliated and swim and feed in the planktonic protists.

Locomotion in Sea Cucumber (5 Marks)

Sea cucumbers are mostly sluggish burrowers and creepers, although some swim by undulating their bodies from side to side. • Locomotion using tube feet is inefficient, because the tube feet are not anchored by body wall ossicles. “Locomotion more commonly results from contractions of bodywall muscles that produce wormlike, locomotor waves that pass along the length of the body.

One way that sea cucumbers can confuse or harm predators is by propelling their own toxic internal organs from their bodies in the direction of an attacker. • The organs grow back, and it may save them from being eaten.

Role of Insects in pollination (5 marks)

Insects are responsible for the pollination of approximately 65% of all plant species. •Insects and flowering plants have co evolutionary relationships that directly benefit humans. The annual value of insect-pollinated crops is estimated at \$19 billion per year in the United States.

Insect pollination is crucial to most gardens and is as simple as insects like bees, butterflies and wasps flying from flower to flower in order to collect nectar. •In the process, pollen collects on their bodies and rubs off on other flowers that they visit.

Stomach of Starfish (3 Marks)

The mouth opens to a short esophagus and then to a large stomach that fills most of the coelom of the central disk. The stomach is divided into two regions. •The larger, oral stomach, sometimes called the cardiac stomach, receives ingested food. •It joins the smaller, aboral stomach, sometimes called the pyloric stomach.

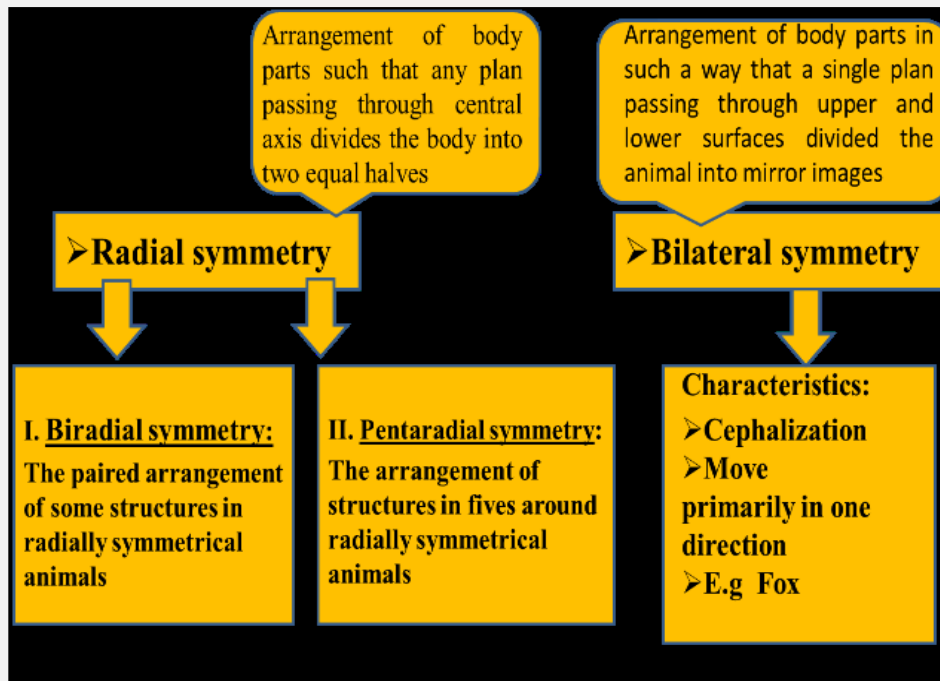
Typhlosole (2 Marks)

A dorsal fold of the luminal epithelium called the typhlosole substantially increases the surface area of the intestine.

Skeleton of Sea Urchins (Phylum Echinodermata)

- Sea urchins are rounded, and their oral end is oriented toward the substrate.
- Their skeleton, called a test, consists of 10 closely fitting plates that arch between oral and aboral ends. Five rows of ambulacral plates have openings for tube feet, and alternate with five inter ambulacral plates, which have tubercles for the articulation of spines.
- The base of each spine is a concave socket, and muscles at its base move the spine. • Spines are often sharp and sometimes hollow, and they may contain venom dangerous to swimmers.

Differentiate b/t Radial & Bilateral Symmetry



Functions of Tube Feet

The water-vascular system of echinoderms is a series of water-filled canals, and their extensions are called **tube feet**.

A tube foot often has a suction cup at its distal end. • When the foot extends and contacts solid substrate, muscles of the suction cup contract and create a vacuum. • In some taxa, tube feet have a pointed or blunt distal end. • These echinoderms may extend their tube feet into a soft substrate to secure contact during locomotion or to sift sediment during feeding.

Tympanal (tympanic) organs (5 Marks)

Tympanal (tympanic) organs are in the legs of crickets and katydids (order Orthoptera), in the abdomen of grasshoppers (order Orthoptera) and some moths (order Lepidoptera), and in the thorax of other moths. • Tympanal organs consist of a thin, cuticular membrane covering a large air sac. • The air sac acts as a resonating chamber.

- Just under the membrane are sensory cells that detect pressure waves. • Grasshopper tympanal organs can detect sounds in the range of 1,000 to 50,000 cps. • The human ear can detect sounds between 20 and 20,000 cps. • Bilateral placement of tympanal organs allows insects to discriminate the direction and origin of a sound

Hemal system in echinoderm (3 marks)

In echinoderms, the blood vascular system is called hemal system, which is poorly developed in star fish and plays little role in circulation. The system consists of small fluid-filled sinus channels

that lack a distinct lining. The haemal channels are actually intercommunicating spaces and are not true blood vessels.

Cnidocytes (2 Marks)

Epidermal and/or gastrodermal cells called cnidocytes produce structures called nematocysts, which are used for attachment, defense and feeding. The cnidocyte has a modified cilium, called a **cnidocil**.

Define vector (2 Marks)

A vector is a living organism that transmits an infectious agent from an infected animal to a human or another animal. Vectors are frequently arthropods, such as mosquitoes, ticks, flies, fleas and lice.

Stages of Complete metamorphosis (2 Marks)

Complete metamorphosis consists of 4 stages. Complete metamorphosis is the type of insect development that includes egg, larva, pupal, and adult stages, which differ greatly in morphology. The lifecycle of butterflies, ants, fleas, bees, beetles, moths, and wasps are examples of the complete metamorphosis.

Sexual Reproduction in Bivalve (3 Marks)

Most bivalves are dioecious. A few are monoecious, and some of these species are protandric. Gonads are in the visceral mass, where they surround the looped intestine. Ducts of these gonads open directly to the mantle cavity or by the nephridiopore to the mantle cavity. Most bivalves exhibit external fertilization. Gametes exit through the suprabranchial chamber of the mantle cavity and the exhalant opening.

What are the Effects of absence of androgenic Hormone on male and female characteristics?? (3 Marks)

- Normally, androgenic hormone(s) promotes the development of testes and male characteristics, such as gonopods
- The removal of androgenic glands from males results in the development of female sex characteristics, and if androgenic glands are experimentally implanted into a female, she develops testes and gonopods

What is Ventilating Mechanism in Insects? (5 Marks)

- Most insects have ventilating mechanisms that move air into and out of the tracheal system.
- Insects do not breathe through their mouths as we do.
- They do not have lungs and their blood, which is a watery, yellowish liquid, does not carry oxygen and carbon dioxide around their bodies.

For example, contracting flight muscles alternatively compress and expand the larger tracheal trunks and thereby ventilate the tracheae • In some insects, carbon dioxide that metabolically active cells produce is sequestered in the hemocoel as bicarbonate ions (HCO_3). • As oxygen diffuses from the tracheae to the body tissues, and is not replaced by carbon dioxide, a vacuum is created that draws more air into the spiracles • This process is called passive suction.

Regeneration in Sea Star (5 Marks)

- Sea stars are well known for their powers of regeneration. • They can regenerate any part of a broken arm. • In a few species, an entire sea star can be regenerated from a broken arm if the arm contains a portion of the central disk.
- A sea star that has the ability to regenerate amputated limbs must first undergo a repair phase to heal the exposed wound. • Once the wound is healed, the sea star can begin to generate new cells, which in turn, sparks new growth. • Cell proliferation that results in the growth of the new limb occurs in the final phase. • If the final phase is interrupted, the resulting new limb may be deformed.

Structure And Function Similarity In Crustaceans and Crayfish (3 Marks)

Crayfish illustrate general crustacean structure and function. • The body of a crayfish is divided into two regions. • A cephalothorax is derived from developmental fusion of a sensory and feeding tagma (the head) with a locomotor tagma (the thorax). • The exoskeleton of the cephalothorax extends laterally and ventrally to form a shield like carapace. • The abdomen is posterior to the cephalothorax, has locomotor and visceral functions, and in crayfish, takes the form of a muscular “tail.”

Spiral Cleavage (2 Marks)

- Holoblastic cleavage that is typical of protostomes. • Spiral cleavage is characterized by arrangement of the blastomeres of each upper tier over the cell junctions of the next lower tier.
- So that the blastomeres spiral around the pole-to-pole axis of the embryo.

Determinate Cleavage (2 Marks)

Determinate cleavage (also called mosaic cleavage) is in most protostomes. • It results in the developmental fate of the cells being set early in the embryo development. • Each blastomere produced by early embryonic cleavage does not have the capacity to develop into a complete embryo.

PTM Stand For?? (2 Marks)

Potato tuber moth (PTM) on Irish Potatoes.

Ectotherms & Heterothermy (3 Marks)

Ectotherms are the animals in which their heat source is primarily from the environment.

Heterothermy ("hetero", different) occurs in animals which are able to switch from endothermy to ectothermy.

Endotherms are the animals which can produce heat internally by biochemical processes.

External fertilization in Bivalves? (2 Marks)

Most bivalves exhibit external fertilization. Gametes exit through the suprabranchial chamber of the mantle cavity and the exhalant opening.

Coelomates Examples?? (2 Marks)

Coelomates include mollusks, arthropods, echinoderms, and chordates.

Central Nervous System of Insects (5 Marks)

The central nervous system consists of a series of ganglia that supply nerves to successive segments of the body. The three main ganglia in the head (protocerebrum, deutocerebrum, and tritocerebrum) commonly are fused to form the brain. The rest of the ganglionic chain lies below the alimentary canal against the ventral body surface.

In most insects the number of separate ganglia has been reduced by fusion. The last abdominal ganglion always serves several segments. In homopterans and heteropterans all the abdominal ganglia usually fuse with mesothoracic and metathoracic ganglia.

If sea urchin is invertebrates Name its Phylum? (2 Marks)

Sea urchin is invertebrates and belong to phylum Echinoderm.

Do insects have memory? (2 Marks)

Research has demonstrated that insects are capable of some learning and have a memory. • For example, bees (order Hymenoptera) instinctively recognize flowerlike objects by their shape and ability to absorb ultraviolet light, which makes the center of the flower appear dark • If a bee is rewarded with nectar and pollen, it learns the odor of the flower. • Bees that feed once at artificially scented feeders choose that odor in 90% of subsequent feeding trials. • Odor is a

very reliable cue for bees because it is more constant than color and shape. • Wind, rain, and herbivores may damage the latter.

Aristotle's Lantern or chewing apparatus in Echinoids

Conical shaped chewing apparatus projecting from mouth. - Complex arrangement of muscles and calcareous teeth constituting about 35 ossicles. - Teeth are extruded to scrape algae and other food from rocks. - Five strong teeth, 40 skeletal ossicles and 60 muscles. - Masticating the food. - Feed on algae, bryozoans, coral polyps, and dead animal remnants.

What are Soil dwelling insect

Many other insects are also beneficial. • Soil-dwelling insects play important roles in aeration, drainage, and turnover of soil, and they promote decay processes. • Other insects serve important roles in food webs. • Soil-dwelling insects are hidden in the soil and hard to detect until the sudden appearance of damage creates the demand for rapid, curative treatment.

Define Ossicles (2 Marks)

Series of overlapping skeletal plates known as ossicles. The echinoderm skeleton consists of a series of calcium carbonate plates called ossicles.

What are the function of Trychosysts? (2 Marks)

Trichosysts are pellicular structures primarily used for protection. • They are rod-like or oval organelles oriented perpendicular to the plasma membrane. • In Paramecium, they have a "golf tree" appearance. • The pellicle can discharge trichosysts, which then remain connected to the body by a sticky thread.

Define Royal jelly (2 Marks)

Royal jelly is a honey bee secretion that is used in the nutrition of larvae, as well as adult queens. It is secreted from the glands in the hypopharynx of nurse bees, and fed to all larvae in the colony, regardless of sex or caste.

What is the diet of Crayfish? (2 Marks)

Crayfish prey upon other invertebrates, eat plant matter, and scavenge dead and dying animals.

Cuverian Tubules in Sea Cucumbers??

Other sea cucumbers can evert tubules of the respiratory tree, called Cuverian tubules, through the anus. • These tubules contain sticky secretions and toxins capable of entangling and immobilizing predators.

What is Aurelia?? (2 Marks)

- Aurelia is a plankton feeder. Aurelia is a common scyphozoan in both Pacific and Atlantic coastal waters of North America.
- The margin of its medusa has a fringe of short tentacles and is divided by notches.

The mouth of Aurelia leads to a stomach with four gastric pouches, which contain cnidocyte-laden gastric filaments.

- Radial canals lead from gastric pouches to the margin of the bell.
- In Aurelia, but not all scyphozoans, the canal system is extensively branched and leads to a ring canal around the margin of the medusa.

Alarm pheromones (2 Marks)

Alarm pheromones are defined as chemical substances, produced and released by an organism that warn or alert another of the same species of impending danger. It is provide the Alarm signals visual or auditory.

Example: A sting from one bee alarms other bees in the area, who are likely to attack.

Circulatory system of Insect (2 Marks)

Or

How insect's circulatory system differs from other arthropods?

The circulatory system of insects is similar to that described for other arthropods, although the blood vessels are less well developed. Insects, like all other arthropods, have an open circulatory system which differs in both structure and function from the closed circulatory system found in humans and other vertebrates. In a closed system, blood is always contained within vessels (arteries, veins, capillaries, or the heart itself).

Name the Phylum & Class of Starfish?

Kingdom: Animalia, Phylum: Echinodermata, Class: Asteroidea

Write Scientific name of HUMAN? (2 Marks)

Homo sapiens

Asynchronous Flight (2 Marks)

- Other insects use an indirect or asynchronous flight mechanism.
- Muscles act to change the shape of the exoskeleton for both upward and downward wing strokes.
- Dorsoventral muscles pulling the dorsal exoskeleton (tergum) downward produce the upward wing thrust

Hemimetabolous Metamorphosis (5 Marks)

Some zoologists use an additional classification for insects that have a series of gradual changes in their development, but whose immature form is much different from the adult form usually due to the presence of gills (e.g., mayflies, order Ephemeroptera; dragonflies, order Odonata). • This kind of development is called hemimetabolous (Gr. hemi, half) metamorphosis, and the immatures are aquatic and called naiads (L. naiad, water nymph).

- Insects that undergo hemimetabolous metamorphosis emerge from eggs into nymphs (called larvae by some entomologists) that are similar in shape to the adults.
- They go through several nymphal stages (called instars) before they undergo a final molt into their adult form.
- Also known as incomplete or simple metamorphosis.
- The nymph, or immature insect, resembles the adult in form and eating habits, differing in size, body proportions, and color pattern.

Respiration or Gaseous Exchange in Crayfish? (2 Marks)

- As described, the gills of a crayfish attach to the bases of some cephalothoracic appendages.
- Gills are in a branchial (gill) chamber, the space between the carapace and the lateral body wall.
- The beating of the scaphognathite of the second maxilla drives water anteriorly through the branchial chamber.
- Oxygen and carbon dioxide are exchanged between blood and water across the gill surfaces, and a respiratory pigment, hemocyanin, carries oxygen in blood plasma.

What is trapping? (2 Marks)

Animal trapping, or simply trapping or gin, is the use of a device to remotely catch an animal. Animals may be trapped for a variety of purposes, including food, the fur trade, hunting, pest control, and wildlife management.

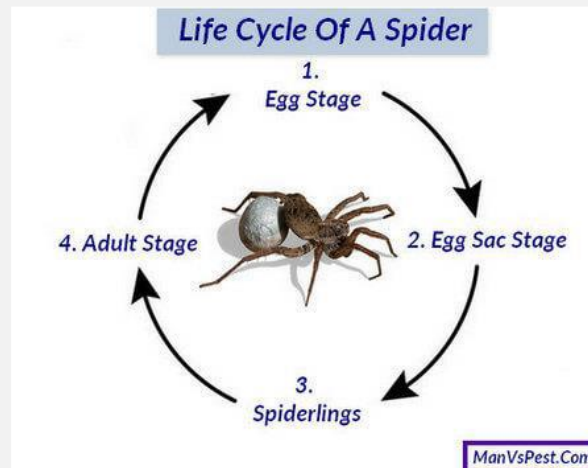
Write Names of 3 Sub-phylum of Phylum Arthropoda (2 Marks)

1. Subphylum Trilobitomorpha (Extinct)
2. Subphylum Chelicerata
3. Subphylum Crustacea
4. Subphylum Uniramia

Four stages of metamorphosis (2 Marks)

- Egg • Larva • pupa

Life cycle Spider? (2



and • Adult.

Stages of a Marks)

Role of Juvenile Hormone in Metamorphosis (2 Marks)

- In immature stages, the corpora allata produces and releases small amounts of juvenile hormone.
- The amount of juvenile hormone circulating in the hemocoel determines the nature of the next molt.
- Large concentrations of juvenile hormone result in a molt to a second immature stage, intermediate concentrations result in a molt to a third immature stage, and low concentrations result in a molt to the adult stage.

Pseudopodia (2 Marks)

Members of the subphylum sarcodina are the amoebae, when feeding and moving, they form temporary cell extensions called pseudopodia (Gr. pseudos means false and podium means little foot).

What is Dermal Branchiae? (3 Marks)

Soft bumps on the body that absorb oxygen from the water are called dermal branchiae.

Thin folds of the body wall, called **dermal branchiae**, extend between **ossicles** and function in gas exchange. • **Papulae**, known as, **Dermal branchiae** or **skin gills**, are projections of the coelom of Asteroidea that serve in respiration and waste removal.

THE TAPEWORMS (3 Marks)

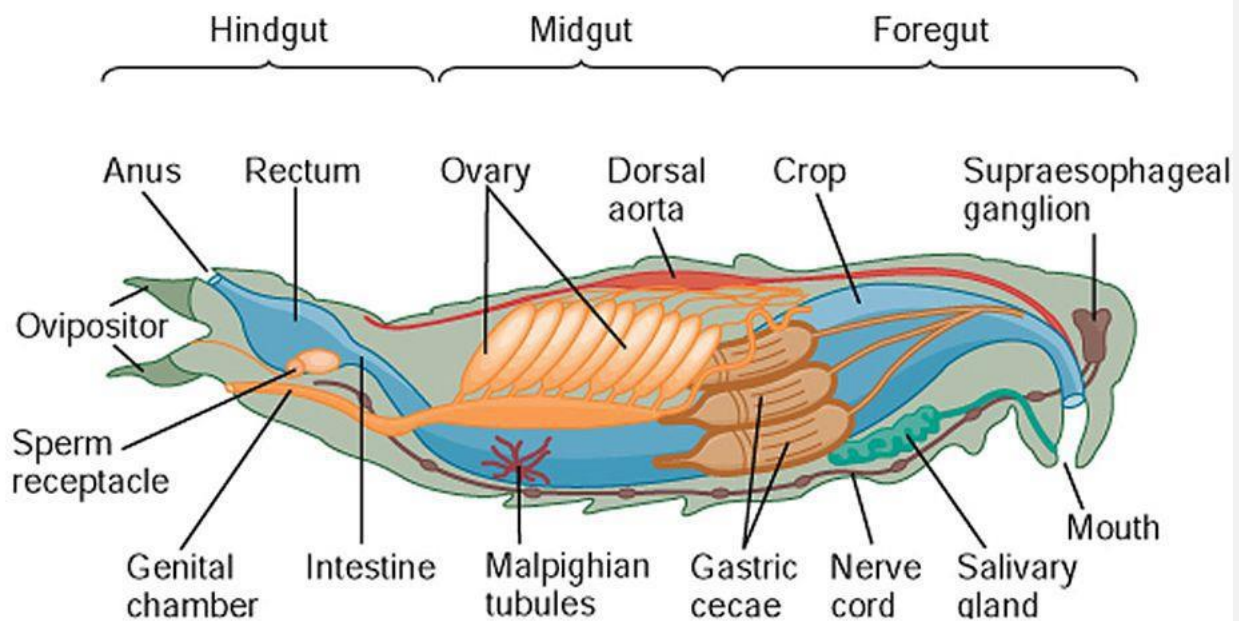
The most highly specialized class of flatworms are members of the class Cestoidea (Gr. keston, girdle eidos, form), commonly called either **tapeworms** or **cestodes**. • All of the approximately 3,500 species are **endoparasites** that usually reside in the vertebrate digestive system.

- Their color is often white with shades of yellow or gray.
 - Adult tapeworms range from **1 mm to 25 m** in length.
 - Two unique adaptations to a parasitic lifestyle characterize tapeworms:
- (1) Tapeworms lack a mouth and digestive tract in all of their life-cycle stages; they absorb nutrients directly across their body wall.
- (2) Most adult tapeworms consist of a long series of repeating units called **proglottids**.

Name of Parts of Insects body (3 Marks)

Internal Structure of a Generalized Insect

Fig. 15.8



Class Hirudinea (2 Marks)

The class Hirudinea (L. hirudin, leech) contains approximately five hundred species of leeches. • Most leeches are freshwater; others are marine or completely terrestrial. • Leeches prey on small invertebrates or feed on the body fluids of vertebrates

Garden centipedes (2 Marks)

Members of the class Symphyla (sim-filah) (Gr. sym, same phyllos, leaf) are small arthropods (2 to 10 mm in length) that occupy soil and leaf mold, superficially resemble centipedes, and are often called garden centipedes.

Why rotifers are called Wheel Animals? (2 Marks)

PHYLUM ROTIFERA (rota, wheel + fera, to bear) derived their name from the characteristic ciliated organ, the corona (Gr. krowe, crown). • Corona is present around lobes on the head. • The cilia of the corona do not beat simultaneously. • Rather each cilium beat earlier than the next cilium. • A wave of beating cilia appears. • It passes around the periphery of the ciliated lobes. • It gives the impression of a pair of spinning wheels. • That is why the rotifers were earlier called wheel animalcules.

Define Nauplius? (2 Marks)

Many other crustaceans have a planktonic, free swimming larva called a **nauplius**

Difference between Brittle Star & Basket Star (2 Marks)

Brittle Star	Basket Star
Brittle stars have unbranched arms, and most have a central disk that ranges in size from 1 to 3 cm.	Basket stars have arms that branch repeatedly.
Brittle stars usually have 5 arms	but Basket star may have up to 10 arms

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Remember me In your Precious Prayers